



The role of structural cues and recency in processing a pre-verbal anaphor in Turkish

Özge Bakay, Faruk Akkuş & Brian Dillon

University of Massachusetts Amherst

Our Study

Principle A: Anaphors such as *each other* in (1) must be c-commanded by their clause-mate antecedents [1].

(1) Linguists_i heard that the neighbors'_j kids_k teased *each other*_{*i/*j/k}.

Question: Is the online retrieval of an antecedent guided by c-command?

Answer: Yes, but with some interference from structurally inaccessible, recent distractors.

Background

There is a lot of evidence that comprehenders rapidly activate the structurally accessible target, with some interference from inaccessible distractors [2-4,6,7, cf.8], especially when targets are less recent in memory [9,cf.10].

As in (2), the experimental designs typically

- investigate post-verbal anaphors [2-4, cf.5] and/or
- distinguish target and distractor using locality (within or outside a clause) rather than c-command [2-10, cf.11].

(2) The bodybuilder_i who worked with the trainer_j injured **himself**_{i/*j}.

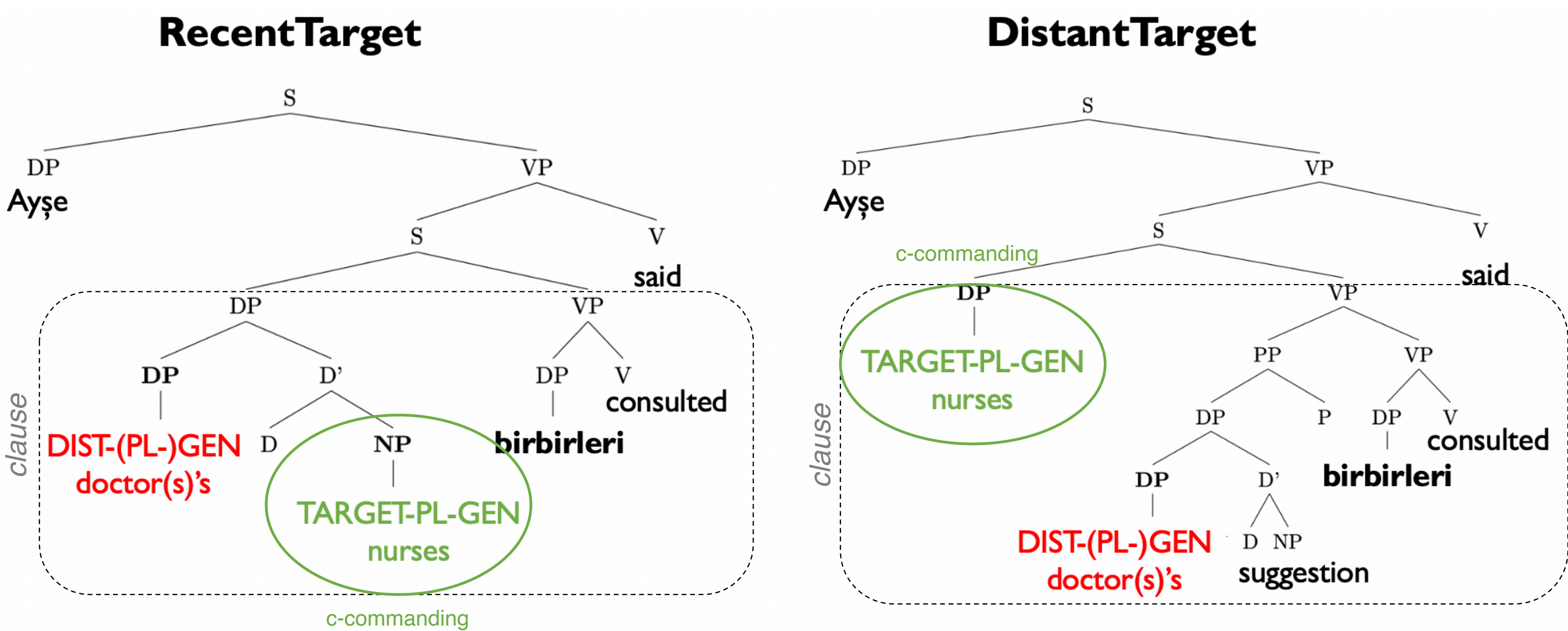
→ **The present study** specifically tests c-command with a pre-verbal anaphor in Turkish.

Stimuli

2x2 design with a c-commanding Target & a non-c-commanding Distractor that are

- in the same clause & identical in terms of morphological case marking

-Int, RecentTarget	Ayşe [Dist-SG-GEN Adj Target-PL-GEN Adv birbirleri Verb] Verb
+Int, RecentTarget	Ayşe [Dist-PL-GEN Adj Target-PL-GEN Adv birbirleri Verb] Verb
-Int, DistantTarget	Ayşe [Target-PL-GEN Dist-SG-GEN N Adv birbirleri Verb] Verb
+Int, DistantTarget	Ayşe [Target-PL-GEN Dist-PL-GEN N Adv birbirleri Verb] Verb



Ayşe said that doctor(s)_i (young) nurses_k (often) consulted each other_{*i/k}.

Ayşe said that nurses_k (often) consulted each other_{*i/k} with doctor(s)_i suggestion.

Offline Antecedent Selection Task

What is the interpretation of *birbirleri* in offline measures?

Exp.1 (N = 32): choosing the antecedent(s) of *birbirleri*

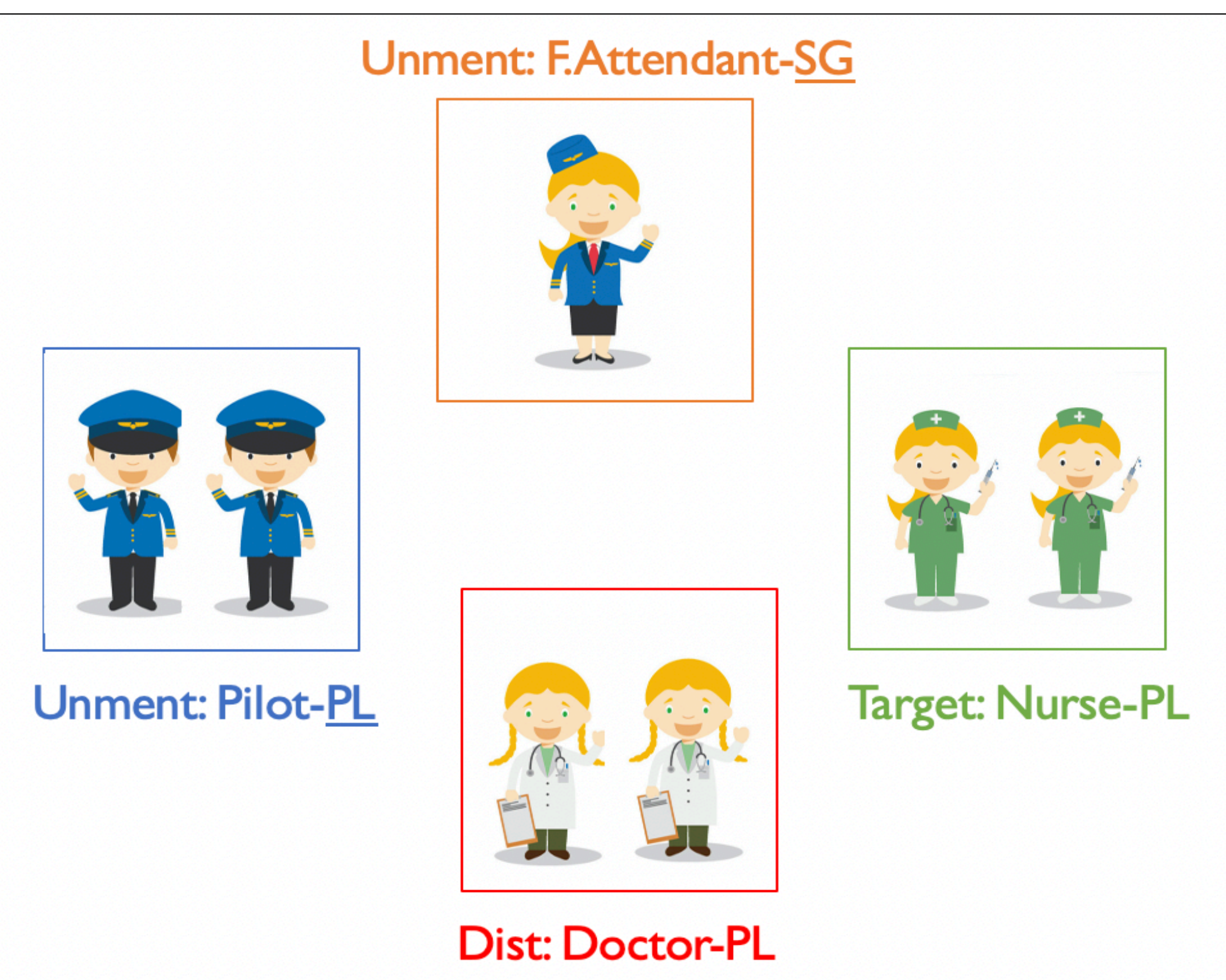
- c-commanding antecedents chosen above 97% of the time,
- more choices of Distractor in +INT than in -INT ($p = .01$) (see [6] for similar findings)

Visual World Experiment

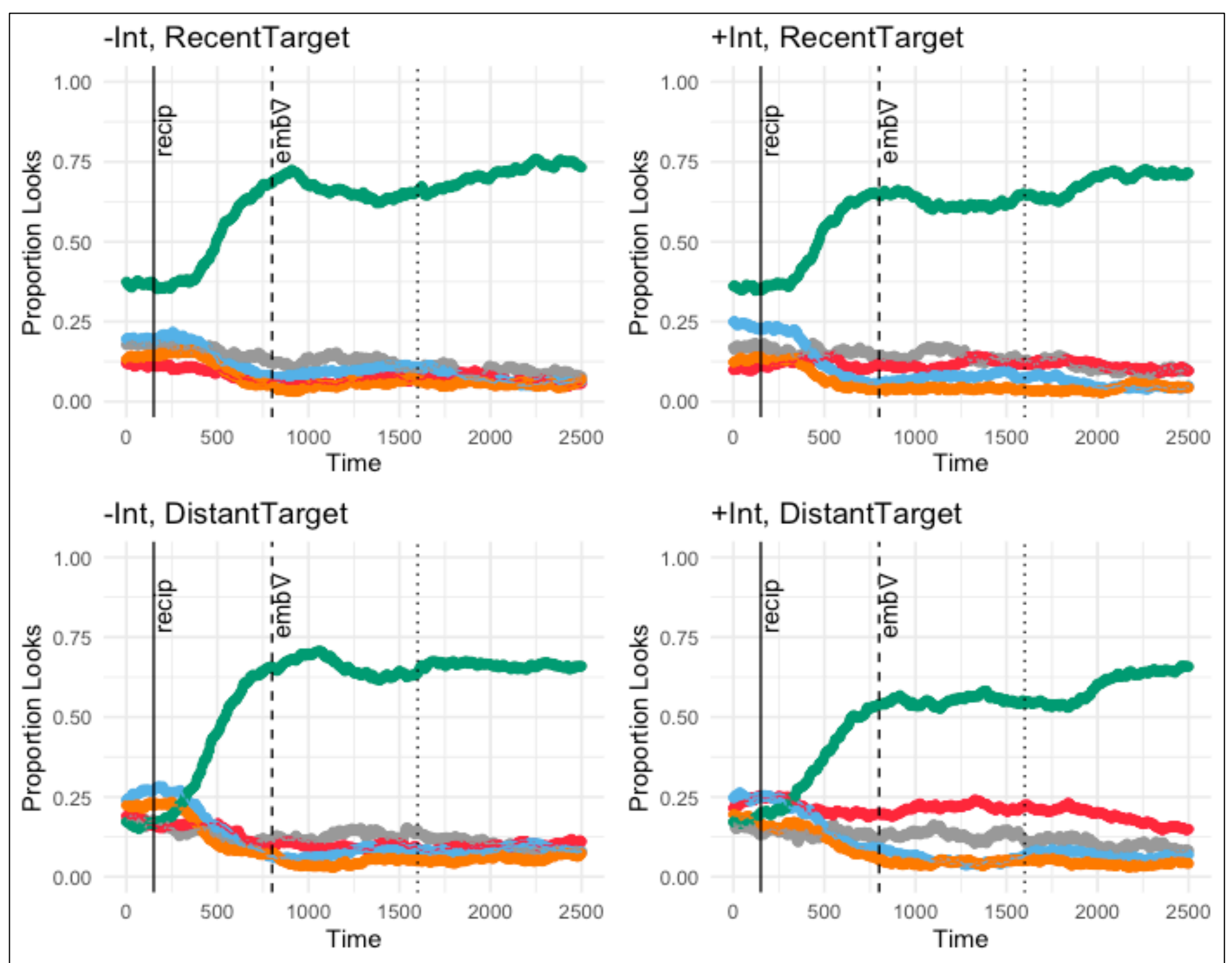
Do listeners use c-command as a cue to the target antecedents in real-time?

Exp.2 (N = 68): listening to statements with the local reciprocal *birbirleri*

Task: clicking on the antecedent that is referred to by *birbirleri*



Example display



Saccade Analysis

How often do listeners launch a saccade towards the distractor [12]?

- *embedded verb window:* overall more saccades towards Distractor in +INT than in -INT ($p < .05$)

Fixation Analysis

$\ln(P(\text{Target}) / (P(\text{UnmPL}))) - \ln(P(\text{Distractor}) / (P(\text{Unm})))$

- overall, more looks to Target than Distractor (p 's $< .001$)
- *embedded verb window:* in DISTTARGET, more looks to Distractor in +INT than -INT ($p < .01$), but no such difference in RECTARGET ($p = .30$)

Conclusion

- *C-command* as a relational cue guides online antecedent retrieval, but distractors may interfere, especially in later stages [10], suggesting that
- both structural and non-structural cues may be used, though with a privileged access for structurally-defined targets [3,5-6,8, cf.2]
- Interference may be reduced when the target is more *recent in memory*, which may be attributed to decreased activations of distant distractors [13].

References: [1] Chomsky, 1981 [2] Runner & Head, 2014 [3] Parker & Phillips, 2017 [4] Patil et al., 2016 [5] Kush & Phillips, 2014 [6] Han et al., 2021 [7] Clackson & Heyer, 2014 [8] Dillon et al., 2013 [9] Cunnings & Felser, 2013 [10] Sturt, 2003 [11] Badecker & Straub, 2002 [12] Kingston et al., 2016 [13] Lewis & Vasishth, 2005 **Acknowledgments:** Thanks to the BUMASA reading group and UMass Psycholinguistics workshop for helpful discussion. Research supported by NSF-BCS 1941485 awarded to BD.

